Experimental Verification of Boyle's Law:

Isothermal Compression of an Ideal Gas

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Abstract

We present an experimental verification of Boyle's law (1662), which states that the pressu

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1. Introduction

Boyle's law, discovered by Robert Boyle in 1662 [1], is one of the foundational principles of therm

$$p \cdot V = C$$

where C is a constant that depends on the amount of gas and the temperature. Equivalently, p = C

The law can be derived from the ideal gas equation pV = nRT, where n is the number of moles, R i

In this work, we experimentally verify Boyle's law using a set of 12 measurements of pressure and

2. Methodology

2.1 Data Collection
The dataset consists of 12 measurements of volume V (in liters) and pressure p (in kPa) taken at c

2.2 Outlier Detection
We employ the Median Absolute Deviation (MAD) method. A measurement is flagged as an outlier

2.3 Least-Squares Fitting
To fit p = C/V, we rewrite as p = C · (1/V). Letting xi = 1/Vi and yi = pi, we minimize S(C) = Σ(xiC

$$C^* = \Sigma x_i y_i \,/\, \Sigma x_i^2$$

Convexity is guaranteed since S''(C) = $2\Sigma x_i^2 \geq 0$.

3. Results

Table 1. Summary statistics of the pV product (excluding outliers).

Number of measurements	12	---
Outliers detected	1	---
Mean pV	249.3	kPa·L
Std dev pV	2.6	kPa·L
Median pV	248.4	kPa·L
Min pV	246.4	kPa·L
Max pV	253.4	kPa·L
Relative std dev	1.06	%
Fitted C	248.9	kPa·L
R^2 of fit	0.9993	---
Temperature	293.15	K

3.2 Outlier Analysis

One measurement at V = 2.20 L, p = 1.136 kPa is flagged as an outlier. The expected pressure is

3.3 Goodness of Fit

The least-squares fit yields C* = 248.9 kPa·L with R^2 = 0.9993. This indicates that the inverse pro

4. Discussion

The experimental results strongly support Boyle's law. The relative standard deviation of the pV p

The fitted constant C = 248.9 kPa·L can be compared with the ideal gas prediction C = nRT. Using

5. Formal Verification
All mathematical foundations have been verified in Lean 4 with Mathlib:
• Least-squares derivative: $S'(c) = 2(c\Sigma x^2 - \Sigma xy)$
• Normal equation: $S'(C^*) = 0$ at $C^* = \Sigma xy/\Sigma x^2$
• Convexity: $S''(c) = 2\Sigma x^2 \geq 0$
• Boyle's law: $p \cdot V = C$, $V \neq 0 \Longrightarrow p = C/V$

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